

Course No. 61102

Name : Sophia LaSala

**State University of New York
Farmingdale State College**

**Department of
Electrical and Computer Engineering Technology**

Experiment # 11

Title: The Common Emitter Amplifier

Date of Experiment : 7/13/2026

Lab Partners :

Date Submitted : 7/19/2026

Marc Ramirez

Date Due : 7/19/2026

Objective

The following experiment focuses on a Common Emitter Amplifier and its typical behaviors; the lab analyzes the circuit in both AC and DC analysis.

Equipment

- 1 - Fluke 8808A Digital Multi-Meter
- 1 - Oscilloscope
- 1 - Function Generator
- 1 - DC Power Supply
- 1 - Amplifier Test Board
- 3 - Set of BNCs to Alligator Leads
- 1 - Decade Box
- 1 - Set of Meter Leads
- 1 - Ti-89 Titanium Calculator

Conclusion

Upon completion of the experiment large discrepancies ranging from 15.81% to 78.34% were seen throughout the experiment, specifically in the AC analysis sections. The likely cause of these differences is due to the measurements from the oscilloscope. It is possible that the peak value was being measured instead of the peak-to-peak value, which would explain the almost 2 times difference between the vout values in Table #3. Additionally, it is possible that the measurement tool featured on the oscilloscope was having difficulties calculating the peak-to-peak voltages. To ensure this was not the case, in future labs all measurements should be verified using the cursor function. For instance, in this lab the positive and negative voltage magnitudes should have been measured to properly ensure the peak-to-peak voltage amounts. Despite any discrepancies within the data, the figures presented within the lab display the overall trends that would be seen in both theoretical and experimental data.

Overall, the experiment focused on a common emitter amplifier and its theoretical and experimental AC and DC analysis. Understanding how to practically analyze an amplifier is crucial to electrical engineering, as it prepares for larger amplifier circuit designs and creation.

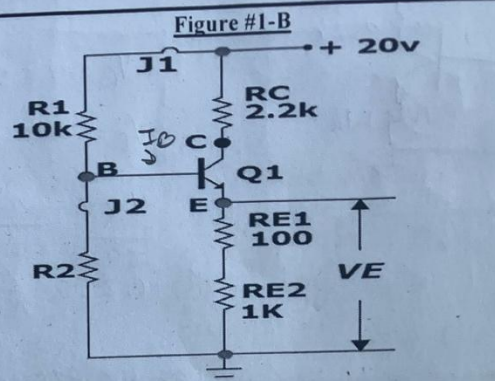
Data Sheet & Theoretical Calculations

EET-118- LABORATORY

EXPERIMENT #10A

TABLE #1 : DC PARAMETERS

Parameter	Nominal	Measured	% Error
R1	9.994k Ω	10k Ω	0.064%
R2*	3.287k Ω	3.07k Ω	7.07%
RC	2.259k Ω	2.2k Ω	2.274%
RE1	100.71 Ω	100 Ω	0.714%
RE2	0.978k Ω	1k Ω	2.24%
Rx	9.968k Ω	10k Ω	0.321%
RL	2.238k Ω	2.2k Ω	1.591%



*Note: R2 will be measured at the end of this part of the experiment.

DC Measurements

Parameter	Measured Values	Computations for IB, hFE, and r'e
VCC	20V	$I_B = I_1 - I_2 = 1.532mA - 1.490mA = 42\mu A$
VC	11.692V	
VE	4.0V	$\beta_{DC} = \frac{I_C}{I_B} = \frac{3.667mA}{42\mu A} = 87.31$
VCE	7.695V	
VB	4.667V	$r'_e = \frac{25mV}{I_E} = \frac{25mV}{3.708mA} = 6.742\Omega$
VBE	0.668V	
IDC	5.182mA	
IC	3.667mA	
I1	1.532mA	
I2	1.490mA	

Results Obtained from Measured Values

IB	42 μA
hFE	87.31
r'e	6.742 Ω

Name	Sophia LeDala	Date	7/13/26	Initial:	
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Table #3-a : Signal Node Voltages

Parameter	Experimental Results Part B – Steps #7, 8 & 9		Theoretical Results: Part F Step #1-c volts p-p	% Error
	Volts p-p	Volts RMS		
voutput	3.36V _{p-p}	1.12V _{rms}	7.02V _{p-p}	68.93%
vc	3.36V _{p-p}	11.718V _{rms}	7.02V _{p-p}	68.93%
ve	80mV	3.996V _{rms}	0.021V _{p-p}	78.34%
vb	480mV	4.666V _{rms}	0.335V _{p-p}	43.28%
vininput	480mV	113.10mV _{rms}	0.335V _{p-p}	43.28%
Es	2.04V	0.705V	2.04V _{p-p}	0%
voutput-nl	6.6V	6.035V	6.65V _{p-p}	0.25%

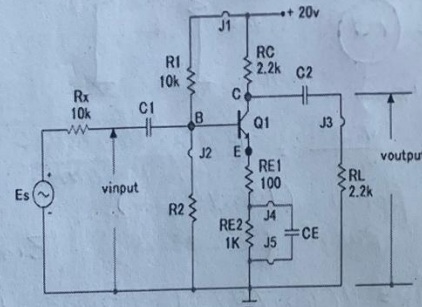
Figure #1-A
Common Emitter Amplifier

	Table #3-b	Table #3-c	
Parameter	Experimental Results Part E	Theoretical Results Part F – Steps #1-b,d,e,f	% Error Part F Step #2
Av	-7	-20.94	66.57%
Av dB	16.90dB	-26.42dB	36.03%
Zinput	1.95k	1.95k	0%
AD	0.24	0.164	46.34%
AE	-1.65	-3.43	51.90%
Av-no-load	-13.85	-19.70	29.70%
ILoad	1.5003mA	1.57mA	6.7%
i-input	123.08μA	86μA	43.12%
Ai	-6.20	-18.270	66.06%
Ai dB	15.85 dB	-25.23dB	37.18%
PLoad	5.05mW	5.51mW	8.85%
PLoad dBm	7.03dBm	7.41dBm	5.13%
Pinput	15.68μW	14.45μW	8.51%
Pinput dBm	-19.05dBm	-18.40dBm	1.9%
Ap	322.07	382.574	15.81%
Ap dB	25.08 dB	25.827dB	2.89%
Zoutput	2.235k	2.235k	0%

Table #2: Signal Waveforms

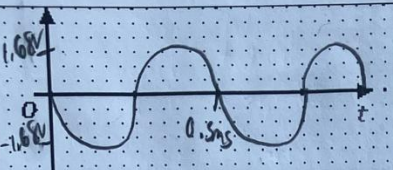
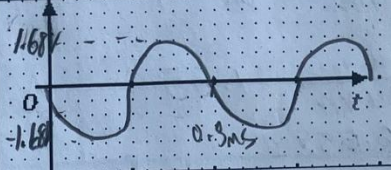
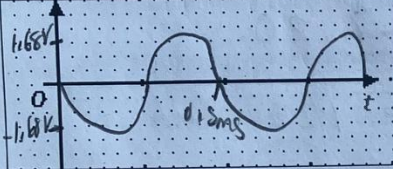
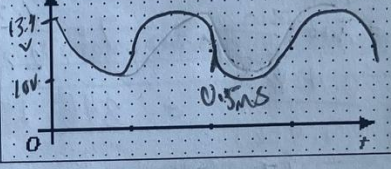
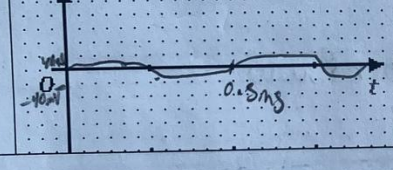
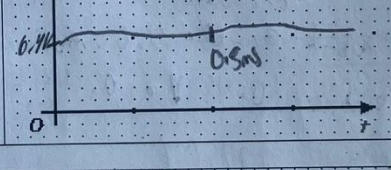
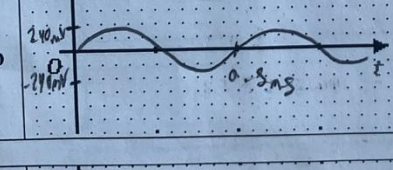
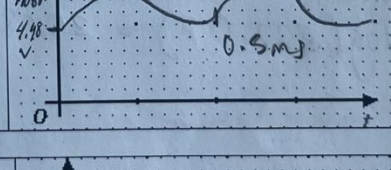
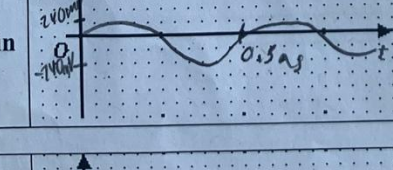
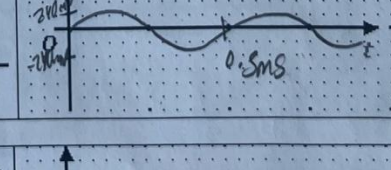
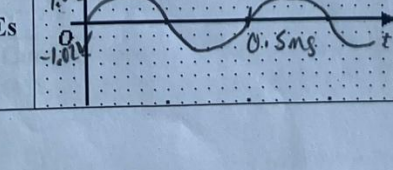
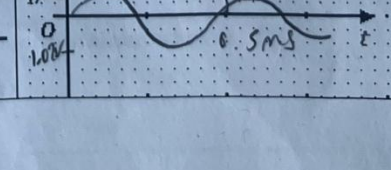
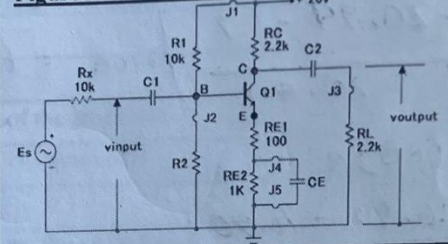
	AC Coupled Oscilloscope	DC Coupled Oscilloscope
vl	 vl p-p = <u>3.36V</u>	 vl p-p = <u>3.36V</u> DC Level = <u>0V</u>
vc	 vc p-p = <u>3.36V</u>	 vc p-p = <u>3.8V</u> DC Level = <u>0.32V</u>
ve	 ve p-p = <u>0.8mV</u>	 ve p-p = <u>200mV</u> DC Level = <u>6.4V</u>
vb	 vb p-p = <u>480mV</u>	 vb p-p = <u>480mV</u> DC Level = <u>4.24V</u>
vin	 vin p-p = <u>480mV</u>	 vin p-p = <u>480mV</u> DC Level = <u>0</u>
Es	 Es p-p = <u>2.04V</u>	 Es p-p = <u>2.04V</u> DC Level = <u>0V</u>

Table #4

Parameter	Experimental Results Part B Volts p-p	Theoretical Results: Part E volts p-p	% Error
voutput	3.40Vpp	7.02Vpp	68.93%
vc	3.40Vpp	7.02Vpp	68.93%
ve	440mVpp	0.021Vpp	78.34%
vce	4Vpp	7Vpp	42.86%

Figure #1-A: Common Emitter Amplifier



Verify you have accounted for the phase relationship that exists between each signal and Es.

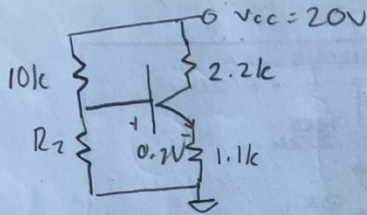
AC Coupled Oscilloscope Measurements		DC Coupled Oscilloscope Measurements	
Es		Es p-p = 2.2V	Es p-p = 2.2V DC Level = 0V
vc		vc p-p = 3.4V	vc p-p = 3.4V DC Level = 11.7V
ve		ve p-p = 440mV	ve p-p = 600mV DC Level = 4V
vce		vce p-p = 3.6V	vce p-p = 4V DC Level = 7.2V

Computation for VCE p-p from measure values obtained above:

$$3.40Vpp - 0.4Vpp = 3.0Vpp$$

Table # 1 Calculations

$$V_E = 4V \quad V_{BE} = 0.7V$$



$$V_B = 4V + 0.7V = 4.7V$$

$$V_B = V_{cc} \cdot \frac{R_2}{R_1 + R_2}$$

$$4.7V = \frac{20V \cdot R_2}{10 + R_2}$$

$$(10 + R_2) 0.235 = \frac{R_2}{10 + R_2} (10 + R_2)$$

$$\frac{2.35}{-0.235} + \frac{0.235 R_2}{-0.235} = \frac{R_2}{-0.235}$$

$$\frac{2.35}{0.765} = \frac{0.765 R_2}{0.765} \quad R_2 = 3.07k\Omega$$

% Error

R_1 : $[k\Omega]$

$$\left| \frac{10 - 9.994}{10} \right| \cdot 100 = 0.06\%$$

R_2 : $[k\Omega]$

$$\left| \frac{3.07 - 3.287}{3.07} \right| \cdot 100 = 7.07\%$$

R_{E1} : $[k\Omega]$

$$\left| \frac{100 - 100.71}{100} \right| \cdot 100 = 0.71\%$$

R_{E2} : $[k\Omega]$

$$\left| \frac{1 - 0.978}{1} \right| \cdot 100 = 2.2\%$$

R_C : $[k\Omega]$

$$\left| \frac{2.2 - 2.25}{2.2} \right| \cdot 100 = 2.27\%$$

R_x : $[k\Omega]$

$$\left| \frac{10 - 9.968}{10} \right| \cdot 100 = 0.32\%$$

R_L : $[k\Omega]$

$$\left| \frac{2.2 - 2.235}{2.2} \right| \cdot 100 = 1.59\%$$

% Error:

A_v :

$$\left| \frac{-20.94 - -7}{-20.94} \right| \cdot 100 = 66.57\%$$

A_v (dB):

$$\left| \frac{26.42 - 16.90}{26.42} \right| \cdot 100 = 36.03\%$$

A_D :

$$\left| \frac{0.164 - 0.24}{0.164} \right| \cdot 100 = 46.34\%$$

A_e :

$$\left| \frac{-3.43 + 1.65}{3.43} \right| \cdot 100 = 51.90\%$$

A_v (nd):

$$\left| \frac{-19.70 + 13.85}{19.70} \right| \cdot 100 = 29.70\%$$

I_{Load} (mA)

$$\left| \frac{1.57 - 1.503}{1.57} \right| \cdot 100 = 6.74\%$$

I_{input} (mA)

$$\left| \frac{86 - 123.08}{86} \right| \cdot 100 = 43.12\%$$

A_i :

$$\left| \frac{-18.270 + 6.20}{-18.270} \right| \cdot 100 = 66.06\%$$

A_i (dB)

$$\left| \frac{25.23 - 15.85}{25.23} \right| \cdot 100 = 37.18\%$$

P_{Load} (mW)

$$\left| \frac{5.51 - 5.05}{5.51} \right| \cdot 100 = 8.35\%$$

P_{Load} (dBm):

$$\left| \frac{7.41 - 7.03}{7.41} \right| \cdot 100 = 5.13\%$$

P_{input} (mW)

$$\left| \frac{14.45 - 15.68}{14.45} \right| \cdot 100 = 8.51\%$$

P_{input} (dB):

$$\left| \frac{-18.40 + 18.05}{-18.40} \right| \cdot 100 = 1.9\%$$

A_p :

$$\left| \frac{382.57 - 322.07}{382.57} \right| \cdot 100 = 15.81\%$$

A_p (dB):

$$\left| \frac{25.827 - 25.08}{25.827} \right| \cdot 100 = 2.89\%$$

V_{out}/V_c (V/V)

$$\left| \frac{3.36 - 7.02}{3.36} \right| \cdot 100 = 68.93\%$$

V_e (V/V)

$$\left| \frac{21 - 80}{21} \right| \cdot 100 = 78.34\%$$

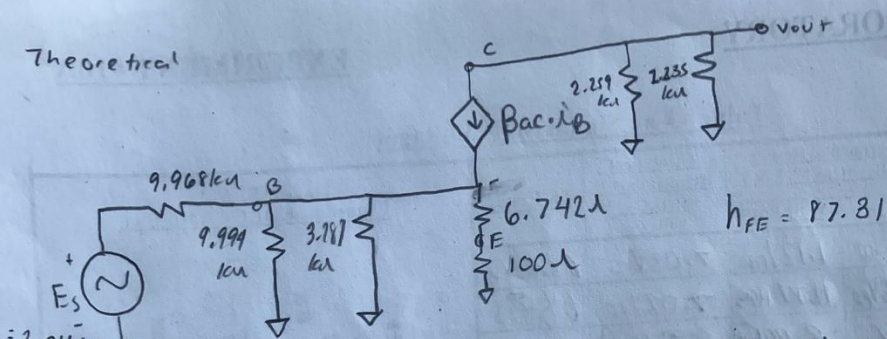
V_b (V/V)

$$\left| \frac{335 - 480}{335} \right| \cdot 100 = 43.28\%$$

V_{out} (V/V)

$$\left| \frac{6.65 - 6.6}{6.65} \right| \cdot 100 = 0.75\%$$

Theoretical



$$A_v = \frac{v_{out}}{v_{in}} = \frac{-\beta_{ac} i_B (2.235 \text{ k}\Omega)}{\beta_{ac} i_B (106.742 \text{ }\Omega)} = -20.94 \quad A_v(\text{dB}) = 20 \log(20.94) = -26.3$$

$$v_{in} = \beta_{ac} i_B (106.742 \text{ }\Omega) \quad Z_{in} = 9.994 \text{ k}\Omega // 3.287 \text{ k}\Omega // r'_b$$

$$v_{out} = -\beta_{ac} i_B (2.235 \text{ k}\Omega) \quad r'_b = 106.742 \text{ }\Omega \cdot \beta_{ac} = 106.742 \text{ }\Omega \cdot 87.81 = 9.32 \text{ k}\Omega$$

$$Z_{in} = 1.95 \text{ k}\Omega$$

$$(A_v)_{NL} = \frac{0.6 \text{ V}_{pp}}{0.335} = 19.7$$

$$A_D = \frac{v_{in}}{E_s} = \frac{E_s \cdot \frac{1.95 \text{ k}\Omega}{1.95 \text{ k}\Omega + 9.968 \text{ k}\Omega}}{E_s} = 0.164$$

$$A_e = A_v \cdot A_D = -20.94 \cdot 0.164 = -3.43$$

$$A_i = A_v \cdot \frac{Z_{in}}{R_{load}} = -20.94 \cdot \frac{1.95 \text{ k}\Omega}{2.235 \text{ k}\Omega} = -18.27$$

$$A_p = A_v \cdot A_i = -20.94 \cdot -18.27 = 382.57$$

$$A_D = \frac{v_{in}}{E_s} \quad v_{in} = A_D \cdot E_s \rightarrow 2.04 \text{ V}_{pp} \cdot 0.164 = 0.335 \text{ V}_{pp}$$

$$A_v = \frac{v_{out}}{v_{in}} \quad v_{out} = A_v \cdot v_{in} \rightarrow -20.94 \cdot 0.335 \text{ V}_{pp} = 7.015 \text{ V}_{pp}$$

$$i_{in} = \frac{v_{in}}{Z_{in}} = \frac{0.335 \text{ V}_{pp}}{1.95 \text{ k}\Omega} = 0.172 \text{ mA}_{pp} = 86 \text{ }\mu\text{A}$$

$$i_{load} = \frac{v_{out}}{R_{load}} = 1.57 \text{ mA}$$

$$P_{load} = v_{out} \cdot i_{load} = 3.51 \text{ V} \cdot 1.57 \text{ mA} = 5.51 \text{ mW} \quad P_{load}(\text{dBm}) = 10 \log\left(\frac{5.51 \text{ mW}}{1 \text{ mW}}\right)$$

$$P_{in} = v_{in} \cdot i_{in} = 0.168 \text{ V} \cdot 0.086 \text{ mA} = 14.45 \text{ }\mu\text{W} \quad P_{in}(\text{dBm}) = 10 \log\left(\frac{14.45 \text{ }\mu\text{W}}{1 \text{ mW}}\right)$$

$$v_b = v_{in} \quad v_o = v_b \cdot \frac{6.742}{106.742} = 0.335 \cdot \frac{6.742}{106.742} = 0.021 \text{ V}_{pp}$$

measured

$$A_v = \frac{3.36V_{pp}}{0.480V_{pp}} = 7$$

$$A_v(dB) = 20 \log(7) = 16.90$$

$$A_D = \frac{0.480V_{pp}}{2.04V_{pp}} = 0.24$$

$$A_e = \frac{3.36V_{pp}}{2.04V_{pp}} = -1.65$$

$$(A_v)_{nl} = \frac{6.65V_{pp}}{0.480V_{pp}} = 13.85$$

$$A_i = \frac{i_{load}}{i_{in}} = A_v \frac{Z_{in}}{R_{load}} = -7 \cdot \frac{1.75}{2.2} = -6.20$$

$$i_{in} = \frac{v_{in}}{Z_{in}} = \frac{0.48V_{pp}}{1.95k\Omega} = 246.15\mu A_{pp} = 123.08\mu A$$

$$i_{load} = \frac{v_{out}}{R_{load}} = \frac{3.36V_{pp}}{2.235k\Omega} = 1.503mA$$

$$P_{load} = 1.503mA \cdot 3.36V = 5.05mW$$

$$P_{input} = v_{in} \cdot i_{in} = 0.480V \cdot 0.12308mA = 15.68\mu W$$

$$A_p = \frac{5.05mW}{15.68\mu W} = 322.07$$

$$A_p(dB) = 10 \log(322.07) = 25.08dB$$

Data Table

Table #1 DC Parameters

Parameter	Nominal	Measured	% Error
R1	9.994 k Ω	10 k Ω	0.06%
R2	3.287 k Ω	3.07 k Ω	7.07%
RC	2.259 k Ω	2.20 k Ω	2.27%
RE1	100.71 Ω	100 Ω	0.71%
RE2	0.978 k Ω	1 k Ω	2.20%
Rx	9.968 k Ω	10 k Ω	0.32%
RL	2.235 k Ω	2.2 k Ω	1.59%

Parameter	Measured Values
VCC	20 V
VC	11.692 V
VE	4.0 V
VCE	7.695 V
VB	4.667 V
VBE	0.668 V
IDC	5.182 mA
IC	3.667 mA
I1	1.532 mA
I2	1.490 mA

IB	42 μ A
hFE	87.31
r'e	6.742 Ω

Table #2 Signal Waveforms

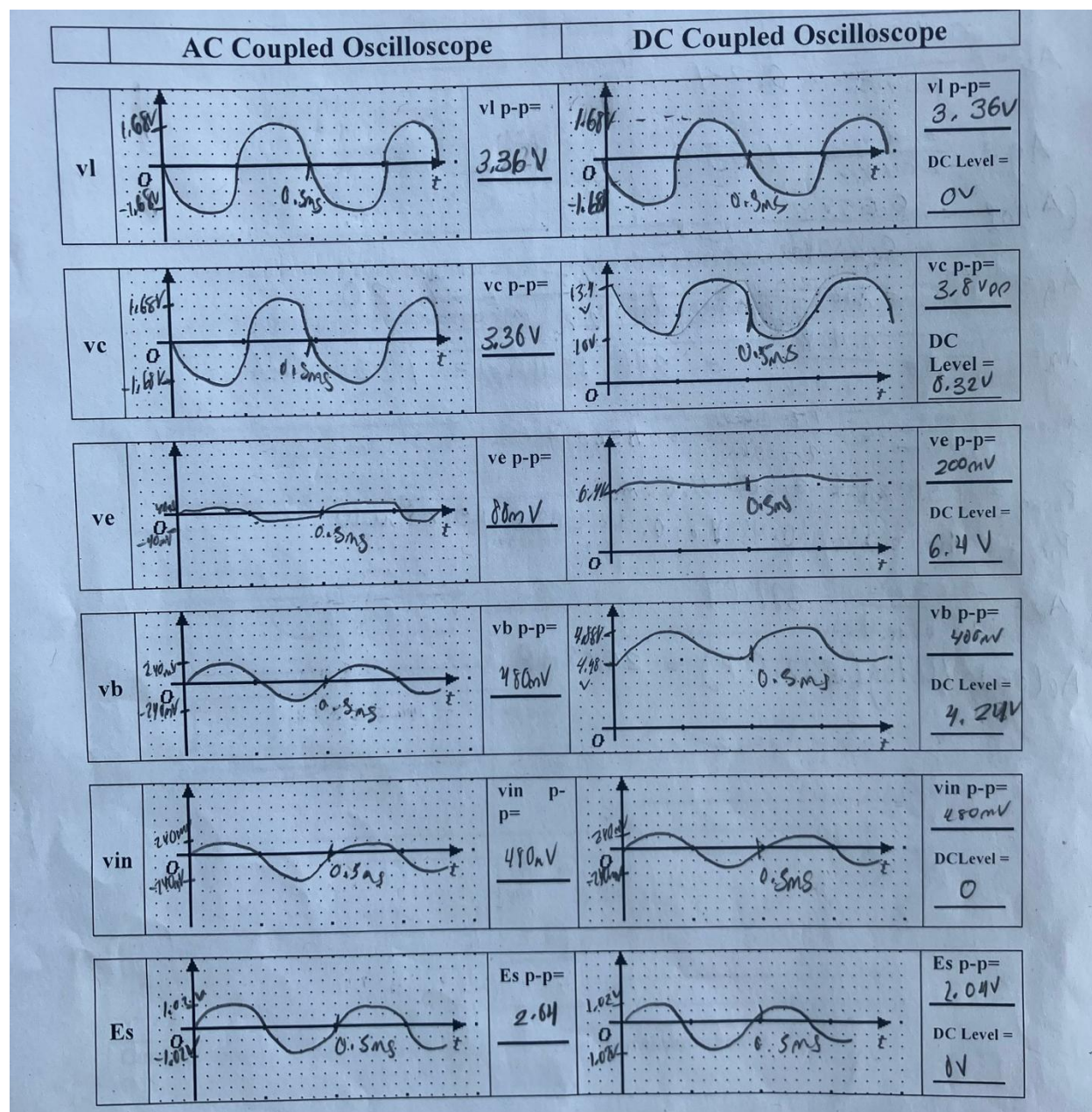


Table #3 Signal Node Voltages

Parameter	Measured Peak to Peak	Measured RMS	Theoretical Peak to Peak	% Error
voutput	3.36 V _{p-p}	1.12 VRMS	7.02 V _{p-p}	68.93%
vc	3.36 V _{p-p}	11.718 VRMS	7.02 V _{p-p}	68.93%
ve	80 mV _{p-p}	3.996 VRMS	21 mV _{p-p}	78.34%
vb	480 mV _{p-p}	4.666 VRMS	335 mV _{p-p}	43.28%
vinput	480 mV _{p-p}	113 mVRMS	335 mV _{p-p}	43.28%
Es	2.04 V _{p-p}	705 mVRMS	2.04 V _{p-p}	0%
Voutput-nl	6.60 V _{p-p}	6.035 VRMS	6.65 V _{p-p}	0.25%

Parameter	Experimental Results	Theoretical Results	% Error
A _v	-7	-20.94	66.57%
A _v dB	16.90 dB	26.42 dB	36.03%
Z _{input}	1.95 k Ω	1.95 k Ω	0%
A _d	0.24	0.164	46.34%
A _e	-1.65	-3.43	51.90%
A _v (nl)	-13.85	-19.70	29.70%
i _{Load}	1.503 mA	1.570 mA	6.70%
i _{-input}	123.08 μ A	86 μ A	43.12%
A _i	-6.20	-18.270	66.06%
A _i dB	15.85 dB	25.23 dB	37.18%
P _{Load}	5.05 mW	5.51 mW	0.35%
P _{Load} dBm	7.03 dBm	7.41 dBm	5.13%
P _{input}	15.68 μ W	14.45 μ W	8.51%
P _{input} dBm	-18.05 dBm	-18.40 dBm	1.9%
A _p	322.07	382.574	15.81%
A _p dB	25.08 dB	25.827 dB	2.89%
Z _{output}	2.235 k Ω	2.235 k Ω	0%

Table #4

Parameter	Experimental Peak to Peak	Theoretical Peak to Peak	% Error
v _{output}	3.40 V _{p-p}	7.02 V _{p-p}	68.93%
v _c	3.40 V _{p-p}	7.02 V _{p-p}	68.93%
v _e	440 mV _{p-p}	21 mV _{p-p}	78.34%
v _{ce}	4.00 V _{p-p}	7.00 V _{p-p}	42.86%

